



Estimation of orange tree production by regression from video segments under uncontrolled conditions

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Abstract

Precision agriculture covers a wide range of information and communications technologies aimed at supporting current challenges in crop estimation, productivity increase, and food security. In particular, crop yield estimation can provide very valuable information on crop distribution, with the aim of optimising logistics and harvest timing. This paper focuses on deep learning-based regression solutions for estimating the number of visible oranges on trees, from real-world crop row videos captured by a camera placed on a farm vehicle. Count predictions based on individual frames were compared with those based on variable size sequences of frames centred on each tree (video segments). The performance of three deep neural networks designed for regression was evaluated in terms of the regression error and the uncertainty of the estimates, and differences were analysed using nonparametric hypothesis testing. Experiments were conducted on a new dataset composed of annotated video segments of orange tree rows acquired under uncontrolled conditions, which has been made publicly available. Results statistically prove the value of considering multiple frames and the feasibility of yield estimation by regression in the wild. These findings are expected to contribute to optimising decision-making in crop resource management. Unlike most efforts so far, which involve counting fruits by detection in tree images usually captured manually, this work explores counting fruits by regression on trees from real-world video data.

Keywords Precision agriculture · Fruit load estimation · Feature pyramid networks · Annotated image dataset

1 Introduction

Precision agriculture (PA) [1] is an emerging applied research field that aims to optimise farm practices, through the adoption of sensor and smart technologies like Internet of Things (IoT) [2] and artificial intelligence (AI) [3–5]. Decision-making in crop management, environmental sustainability by reducing farming footprint, and economic return by boosting competitiveness are areas that offer countless opportunities and can greatly benefit from innovation in this field.

Over the entire crop life cycle, production estimation could be defined as the measurement of plant physiological traits such as the number or count of product units (e.g. fruits, flowers, etc.), or the total amount or weight of product biomass in a given area, volume, or plant [6]. The fruit number is often referred to as *load*, while the fruit weight, to as *yield* [4]. To date, production estimation mostly relied on an experienced human observer, a manual

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task that can be considered inaccurate, time-consuming, tedious, hardly scalable, and observer dependent [7].

The unprecedented progress of the machine learning and computer vision methods in recent years has allowed the automation of very effective counting solutions [8]. Periodic measurements can lead to time series that might provide rich information on the crop evolution as well as on the efficacy of agricultural practices such as irrigation, fertilisation, etc. Besides, knowing the yield spatial distribution can promote more rational resource assignments.

Counting objects in images is a very popular task in the field of computer vision [9]. Most solution methods can be classified into two dominant paradigms: counting by detection [4, 10] and counting by regression [11]. The first one entails a first step in which object locations are elicited, and a second step in which detected objects are counted. The success of this process strongly depends on the detection accuracy, which is usually very sensitive to object density, object occlusions, and changing lighting conditions [12, 13]. Another drawback of counting by detection is the considerable effort required to annotate images by marking objects of interest. However, when counting is the ultimate goal and object locations are not required, counting by regression has proved to be steadily more accurate than counting by detection when density of agricultural objects increases [8]. Regression can potentially deal with more general scenarios, where part of the production might remain fully hidden by branches, leaves, or other product units. In addition, regression would lead to a direct estimation of the total amount of product biomass (e.g. the total weight in kg), assuming that visible production evidence is correlated with the total amount. A weakness of counting by regression, however, could be a greater difficulty in explaining its predictions.

In the case of fruit load estimation, two main scenarios can be stressed: counting the number of *visible* fruits in a single view of a tree [8, 14] and counting the *total* number of fruits on a tree [15–18]. Assuming that visible fruits on a tree canopy image are a representative sample of the entire fruit population, the former can be sufficient to provide reliable fruit distribution statistics, what can support the comparison of production between different crop regions (spatial distribution), different seasons (temporal distribution), and the optimisation of resource assignment in farm activities. The second scenario usually involves images from multiple (two or four) tree views (sides), from which the total load is estimated. This approach seems appropriate for orchards where separation between trees admits a systematic use of equipment moving around them. According to [12], two-side imaging methods usually suffer from fruit load underestimation, while four-side imaging, from fruit load overestimation.

Unlike previous efforts focused on a single image per side, this work addresses the estimation of the number of visible oranges on a tree side, by combining counts obtained by direct regression over multiple crop row video frames. Using multiple frames per tree is expected to provide robustness to inaccuracies in tree detection and to scene complexities such as fruit overlapping, foliage density, illumination conditions, and sensor-to-fruit angle and distance. To our knowledge, no previous work has dealt with fruit count estimation based on multiple video frames.

Experiments are based on a new database composed of four real videos captured by a camera installed in a farm vehicle that moved along crop rows. From each video, frame sequences centred on each tree were extracted. Each tree-centred sequence was annotated with the average of the corresponding series of frame-based manually annotated counts.

The installation of video cameras and edge computing hardware on farm vehicles, instead of, for example, on unmanned aerial vehicles (UAV or drones), is driven by two fundamental advantages: high frequency of use and low cost. Over the course of a year, farmers use their farm vehicles many times to perform different agricultural tasks, including planting, fertilising, spraying, and harvesting. Recording videos of side views of trees each time would entail negligible costs. In addition, handling the equipment involved would not require specific training or regulation. Therefore, our solution can be considered simple, cost-effective, scalable, and easily integrable.

1.1 Contributions

The main contributions of this work are summarised and discussed below.

- *Multi-frame counting strategy.* Unlike the common practice of analysing individual images by trees obtained under controlled conditions, this work proposes a different use case from real-world crop row videos: the aggregation of multiple frame-based predictions for a more robust fruit counting estimate in uncontrolled environments. Multiple frames per tree are expected to provide robustness to inaccuracies in tree detection and to scene complexities. In addition, the method allows for rigorous handling of estimation uncertainties.
- *Counting by regression based on multi-scale feature maps.* A feature pyramid network [19] (FPN) architecture using an EfficientNetB0 [20] model as the backbone was adapted for regression purposes. Since FPN was originally designed for object detection tasks, the proposed solution allows to evaluate the potential of the multi-scale highly semantic feature maps extracted by

FPN in an unconventional task paradigm: regression-based counting of objects in images.

- *New annotated dataset to test smart agriculture solutions.* A new dataset with annotated video segments acquired in a real orchard under uncontrolled conditions was built and published, which is intended to facilitate future experimental benchmarking. In addition to introducing the database, first baseline results of counting by regression and counting by detection are presented, which are expected to boost research on new counting methods. A fruit annotation methodology is also proposed, which may inspire future efforts.
- *A counting by regression study on a new real-world task.* The proposed regression model and two well-known detection models used as benchmarks were evaluated and compared on the new fruit counting scenario. Results provide new evidence to characterise the weaknesses and strengths of both paradigms (counting by regression and counting by detection) in a new task with unique complexities. Analyses and conclusions drawn, together with other comparable evidence reported in previous works, can provide practical guidelines on which paradigm to choose in each operating scenario.

The code¹ and the dataset² are publicly available to encourage further research.

1.2 Structure of the manuscript

The following sections cover the current state of deep counting estimation in PA (Sect. 2), the materials and methods involved (Sect. 3), the results of the experiments conducted (Sect. 4), and a discussion of results (Sect. 5). Finally, conclusions and future research directions are presented in Sect. 6.

2 Related work

For simplicity, this analysis is limited to yield estimation from images using deep neural networks (DNN), with a focus on counting solutions. Unlike conventional image processing based on hand-crafted features, DNN are able to discover complex features on their own and to use them to feed an optimal or near-optimal decision rule usually inferred within a single learning process. More extensive reviews on smart agriculture based on artificial intelligence can be found in [21–23]. Compared to other surveys on precision agriculture, [21, 23] proposed taxonomies driven by machine learning techniques rather than by smart

farming applications. They put special emphasis on advanced ensemble approaches in the context of crop recommendation systems [22], while exploring opportunities to deploy them through interactive Streamlit-powered web applications. A comprehensive review on generative AI in crop recommendation was conducted in [23], mainly including solutions based on variational autoencoders (VAE) and generative adversarial networks (GAN). These techniques are capable of generating synthetic samples that closely mimics real-world data, to promote better generalisation of DNN models in contexts of real data scarcity.

The exploration of related works was divided into two categories considering the nature of the images: aerial acquired by UAV or satellites and ground images acquired by mobile devices or farm vehicles. In the former, depending on the pixel resolution that could be tens of metres per pixel in satellite images, yield is usually estimated by regression in amounts of product biomass since product units are difficult to detect. In the second category, both paradigms, detection and regression, are suitable.

The use of remote sensing images to forecast crop yield is an emerging research area [24]. DNN models learn how to relate spectral and spatial data from hyperspectral images or from vegetation indexes, such as NDVI (normalised difference vegetation index), to ground-truth yield data. A wheat yield prediction model was introduced in [25], where yield estimates in kg/ha based on the leaf area index (LAI) and leaf nitrogen content (LNC) retrieved from UAV hyperspectral imagery achieved a mean absolute percentage error (MAPE) value of 9.36%. A Bayesian neural network model was trained for corn yield prediction from heterogeneous data sources such as time series satellite products, sequential climate observations, soil property maps, and historical corn yield records, leading to corn yield estimates with a MAPE value of 9.56% [26]. Examples of paddy rice mapping can be found in [27, 28].

Despite the significant breakthrough achieved by satellites and drones in PA, the use of detection algorithms such as YOLO [29] on ground images is still the dominant trend probably due to its lower costs, regulations, and need for training. A version of YOLOv4 [30] to count olive fruit fly is proposed in [31]. In [12], YOLOv2, v3, and v4 models were assessed through transfer learning in detecting and counting ripe oranges. YOLOv4 precision and recall in orange detection over test images were 91.23%, 92.8%, respectively. The impact of lighting conditions on detection effectiveness was investigated, as well as the use of two-side and four-side manually acquired imaging in total load estimation. In [32], an AI-based drone system was introduced to detect and count tomato flowers and fruits through YOLOv5 [33]. Class accuracies were 85%, 99%, and 50% for red tomato, green tomato, and flowers, respectively.

¹ <https://github.com/pjsv1990uji/Orange-count—FPN>.

² <https://smartcai.ambimetrics.com/>.

More recent counting solutions can be found in [34–36]. A Mask R-CNN model [40] integrating Swin Transformer [41] as the backbone was introduced in [34] to segment (detect) fruits on a peach orchard. Experiments were performed on a new database with 125 images of four trees and nine bunches acquired at different distances. The average number of peaches per image was 12 and 2.7 in the images focused on trees and bunches, respectively. The detection performance was measured in terms of the mean average precision (mAP), reaching a value of 0.60. Another solution for peach identification and counting was proposed in [35]. A lightweight YOLOv8 model using the MobileNetV3 [42] as its backbone achieved precision and recall values of 0.979 and a mAP of 0.99 on fruit-focused images acquired by a smartphone. YOLOv8s was also the base model in a rice panicle counting study from drone remote sensing images [36]. Using the panicle count along with the thousand-grain weight, number of grains per panicle, and actual yield data, a number of machine learning models were trained to ultimately perform rice yield estimation. Results showed mAP, precision, and recall values of 0.98, 0.98, and 0.968, respectively, for the counting solution, and a R^2 value of 0.84 for the final yield estimation.

In an effort to deploy a real-world solution, a mobile application to assist farmers estimate apple tree yield is presented in [37]. It includes a linear segmentation model with empirically determined coefficients, which classifies pixels based on their values in different colour bands. The performance of counting apples from segmented images was a MAPE value of 8.52%. Experiments relied on two-side images of 40 trees from a single row of a Pink Lady apple tree orchard. The Pearson correlation coefficient between counted apples and the actual number of apples on trees was calculated (0.6), showing the opportunities for estimating the actual yield from the observed yield. This proposal does not involve any machine learning process.

When object locations are not needed, counting by regression can provide more efficient load estimates [8]. Counting by regression learns how to map image features into a final count estimate or into a density map. This paradigm is particular suitable in complex scenes, where high densities of overlapping objects could make detection of every unit (instance) of interest virtually impossible. In [8], two methods based on regression and one based on detection were compared on 12 agricultural datasets for counting fruit and grains. Results proved that counting by regression clearly outperforms counting by detection in images with large numbers of objects and high levels of occlusion, while performs comparably at lower densities. In [38], a regression model based on Inception-ResNet was

trained to directly estimate the number of visible fruits on a tomato plant. Training was carried out on synthetic images, while test involves only real images. The counting accuracy on test images, based on the relative error of the regression results, was 91.03%.

In [39], two strategies of counting plant leaves were proposed: counting by direct regression and counting by detection/density estimation. The first one involves a feature pyramid network [19] designed to estimate leaf counts from multiple image scales and to combine these estimates into an extra DNN output. The second strategy was based on fusing a density map with estimates by direct regression through a network driven by two losses: detection based and regression based. Both methods showed similar performance. Table 1 summarises works with counting approaches.

Count efforts reported so far are based on a single image per tree view and have been mostly focused on controlled scenarios. Of these, a majority has adopted counting by detection over counting by regression. This paper explores a different use case: to count by regression visible oranges on trees from video data acquired by a farm vehicle in the wild. Each tree's load estimate involves a sequence of frames (roughly) centred on that tree, from which a corresponding set of frame-based estimates (measurements) is obtained. The set mean is proposed as the tree load estimate, which is expected to compensate individual deviations and to provide a more accurate prediction. In addition, multiple measurements admit the calculation of uncertainty measures.

3 Materials and methods

A real use case is recreated: (1) a camera on a tractor records videos of rows of orange trees; (2) trees are detected and represented by tree-centric frame sequences, and (3) each tree's visible fruit load is estimated by combining count estimates by frames. Figure 1 depicts a workflow overview.

Since this work focuses on multi-frame load estimation, it is assumed that there exists a method, outside the scope of this work, capable of extracting frame sequences ideally centred on target trees. For the purposes of this work, frame sequences centred on each tree were extracted manually. We hypothesise that it would be enough for a sequence to contain relevant frames not necessarily centred on the target tree. This *loose* detection method is expected to be tolerant to inaccuracies in tree location.

Table 1 Summary of work related to counting solutions

References	Task	Method	Image type	Results	Data usage
[31]	To count olive fruit flies	Counting by detection (YOLOv4)	Images from camera traps	Precision: 0.84 Recall: 0.97	A single random image partition
[12]	To count the number of oranges	Counting by detection (YOLOv4)	Two-side and four-side manually acquired images	Precision: 0.91 Recall: 0.93	A single random image partition
[32]	To count tomato flowers and fruits	Counting by detection (YOLOv5)	Drone images	Precision: 1 Recall: 0.85	A single random image partition
[34]	To count the number of peaches	Counting by detection (MobileNetV3, YOLOv8s)	Fruit-focused images captured by a smartphone	Precision: 0.979 Recall: 0.979 Mean average precision: 0.99	A single random image partition
[35]	To count the number of peaches	Counting by detection (Swin-T, Mask R-CNN)	Tree- and fruit- focused images (12 and 2.7 peaches/image)	Mean average precision: 0.60	A single random image partition
[36]	To count the number of rice panicles	Counting by detection (YOLOv8s)	Drone remote sensing images (various lighting conditions)	Precision: 0.98 Recall: 0.968 Mean average precision: 0.98	A single random image partition
[37]	To count the number of Pink Lady apples	Counting by detection (linear model)	Two-side images of 40 trees from a single row	Relative error: 0.09	No data partition (no training)
[8]	To estimate the number of fruit or grains	Counting by detection Counting by regression	Crop images from 12 agricultural datasets	Mean absolute error in [0, 5]	A single random image partition
[38]	To count the number of tomatoes	Counting by regression (Inception-ResNet)	Synthetic images for training, real images for testing (Google images)	Mean relative error: 0.09	A single random image partition
[39]	To count leaves in tobacco and Arabidopsis plants	Counting by detection, counting by regression (FPN)	Top view images of plants	Mean average precision: 0.95	A single random image partition
This work	To count the number of oranges	Counting by regression (FPN)	Crop row videos from a camera on a farm vehicle in the wild	Absolute error: 2.3 Relative error: 0.14	Multiple video partitions (all trees in a row in the same subset)

3.1 Dataset

A database called ambimetrics orange counting dataset with video segments obtained in orange tree fields has been created as part of this research.

The dataset consists of five videos of rows of orange trees acquired by a camera mounted on a tractor. They are approximately the same length, since row lengths and acquisition technologies were similar. All recordings were taken around noon on a sunny day. The images are in RGB format (24 bits), and their size is 480 x 640 pixels. Table 2 shows a summary of the dataset. The total number of trees is 99 distributed over 5 rows. Figure 2 illustrates a portion

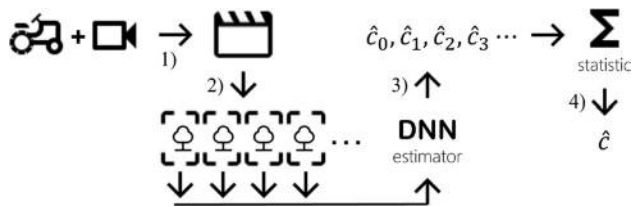


Fig. 1 A conceptual overview: (1) a camera mounted on a tractor records crop row videos; (2) trees are loosely detected and represented by frame sequences; (3) a DNN-based regressor estimates the visible fruit load in each frame; (4) per-frame estimates are summarised by a central tendency statistic

Table 2 Dataset summary: each video's number of trees

Video	No. trees
Row 1	20
Row 2	20
Row 3	19
Row 4	18
Row 5	22
Total	99

of a row as a panoramic image built from a sequence of frames.

One of the videos (row 5 with 22 trees) was strongly affected by sunlight flare, an optical effect that appears in scenes backlit by intense light sources: light spreads out over a wide area of the frame, dark colour tones tend to appear lighter or washed out, while the contrast of the image is noticeably reduced. Since both flare removal [43] and domain shift [44, 45] produced by the alteration of colour distribution are research lines in themselves, the affected data were excluded from experiments. However, we decided to keep it in the dataset to allow advanced

solutions capable of successfully addressing this complexity.

Videos were acquired by a RealSense™ D435 camera mounted on a farm vehicle at a speed between 2 and 5 kms per hour (between 0.83 and 1.39 ms per second), in an orange orchard with distances between neighbouring trees (trunks) of about 3 ms both in the same row and between adjacent rows. The frame rate was set at 30 frames per second (FPS), which allowed capturing enough frames from each tree. Cameras recording each crop row were installed on the far (opposite) side of the vehicle to maximise the distance from the sensor to the trees. The field of view (FOV) was about 2 ms wide by 2.5 ms high. Video frames were extracted using the pyrealsense2 package, a python wrapper for Intel RealSense SDK 2.0.

3.2 Data annotation

Data annotation consists in assigning each sample a piece of information (e.g. a label or a scalar) that represents what needs to be learned and, eventually, predicted. In the counting task, the annotation is the number of visible objects in a scene. In our use case, it is the number of visible oranges on a tree image.

The annotation process is intended to determine a sequence of frames centred on each tree (loose detection), and to annotate both individual frames and the sequence with the number of (clearly) visible oranges. Due to the effort involved in manual labelling, just eleven frames per tree were considered. Sequences were semi-automatically extracted. First, for each tree, a key frame was manually identified; then, segments of $2n + 1$ frames centred on each key frame, with $n \in \mathbb{N}$ ($n = 5$ in this study), were automatically extracted to build the tree-centric sequences (tree representation). Eleven frames were considered sufficient

Fig. 2 Panoramic image from a sequence of video frames



to study the effect of the number of frame-based measurement on the quality of count estimation at tree level. A total of 847 frames (77 trees $\times$ 11 frames/tree) were annotated.

In the process of annotating a frame, only oranges with at least 50% of their surface visible (approximately) were taken into account. It was intended to reduce subjectivity when annotating complex scenes, plenty of overlaps and shadows caused by leaves, branches, and orange clusters. This approach considers the fact that the exact number of oranges is of little interest. However, it allows exploring the feasibility of estimating some yield rate correlated with image data.

Oranges were annotated using the open source tool makesense.ai [46], which supports multiple types of labelling, including points and bounding boxes. The average annotation time per frame using points was roughly between 45 and 60 s depending on the number of oranges present in the frame. The annotated oranges were finally counted by a Python script, and the resulting count assigned as a label to the frame involved.

Figure 3 shows the histogram of frame counts aggregated over all crop rows. As observed, it approaches a Gaussian-shaped distribution. To get a deeper understanding of the annotation process, Fig. 4 shows, by way of example, the frames with the minimum, median, and maximum numbers of oranges annotated.

3.2.1 Annotation by detection: limitations

Automatic annotation of frames with pretrained detection methods was considered, in particular, with the YOLOv5 and YOLOv7 models. They have both been trained to

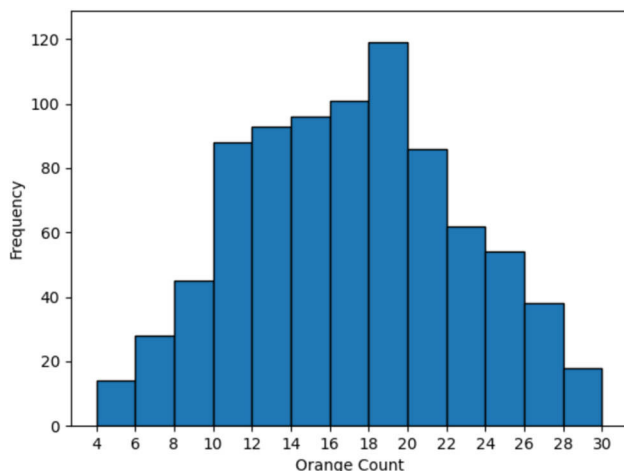


Fig. 3 Histogram of the annotated orange count over the entire dataset

detect oranges from a significant number of diverse samples, so their output can be post-processed to count the number of objects detected. However, exploratory tests showed a number of systematic detection defects; thus, it was decided not to use them. Figure 5 shows examples of detection outputs. First, YOLOv5 tends to overestimate the number of oranges, as it counts some oranges repeatedly and detects oranges in dark regions (with lots of shadows) unsystematically. Second, YOLOv7 hardly detects oranges in heavily shaded regions. Both methods are also affected by the presence of orange clusters (instances are not detected separately) and by oranges on the ground. In the latter case, the system could learn (from the context) to differentiate oranges on a tree (should be counted) from those on the ground (should not be counted).

Despite these limitations, the noisy output of YOLOv5 and YOLOv7 can be considered coarse (and useful) approximations of the target count. Thus, in order to validate manual annotations, correlations between the outputs of YOLOv5 and YOLOv7 (reference values) and manual-driven annotations for each sequence length (see Eq. 3) are calculated as part of the experiments.

3.3 Multi-frame orange counting

The proposed solution focuses on fruit load estimation from tree-centric frame sequences. It assumes that tree detection is performed by a method outside the scope of this work, which determines a predefined-size sequence of frames centred in each target tree. As shown in Fig. 1, each frame is processed separately by a model, and the series of estimates is summarised by a central tendency statistic. Individual estimates are expected to be uncertain; their combination into a tree-level load estimate is expected to be a more balanced and accurate.

More formally, let m be a model fitted to predict (by detection or by regression) the number of visible oranges on a single tree canopy image. Let $x_i = \{f_{ij}\}$ be a frame sequence associated with some tree t_i , composed of l frames f_{ij} with $0 \leq j < l$ and l denoting the sequence length.

The model m can be considered a function that maps an RGB image $I_{h \times w \times 3}$ into a count estimate (a real number),

$$m : I_{h \times w \times 3} \rightarrow \mathbb{R}, \quad (1)$$

where h and w denote the image height and width.

Let $\hat{y}_i = \{\hat{c}_{ij}\}$ be the sequence of count estimates (model outputs) for all frames in x_i such that $\hat{c}_{ij} = m(f_{ij})$. Let S be a statistic of central tendency (e.g. mean, median) that summarises a distribution into a representative value. Then, the estimated count of visible oranges in the tree t_i is computed as follows:

$$\hat{c}_i = S(\hat{y}_i) \quad (2)$$

Fig. 4 Frames with the minimum, median, and maximum number of annotated oranges. Find annotations at the top left corner

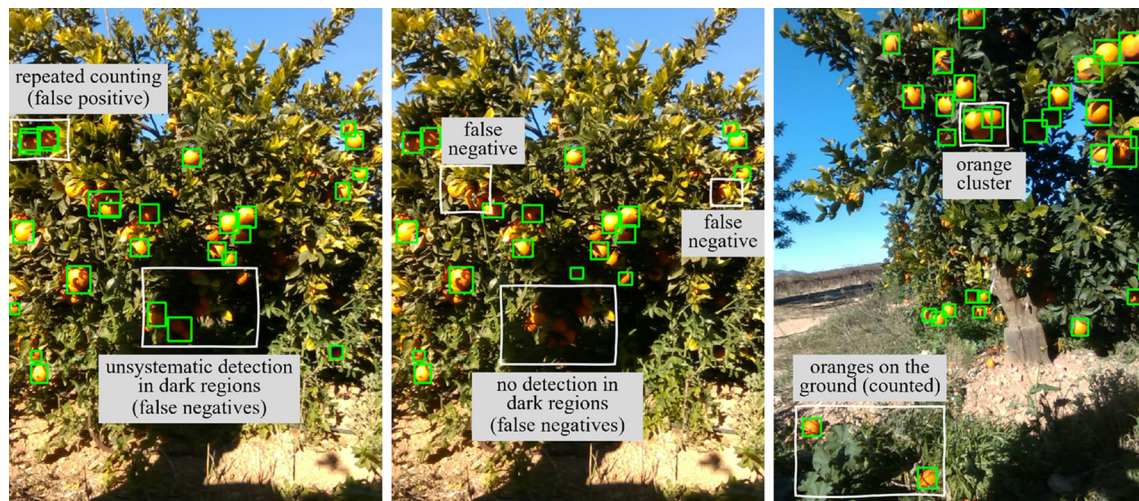
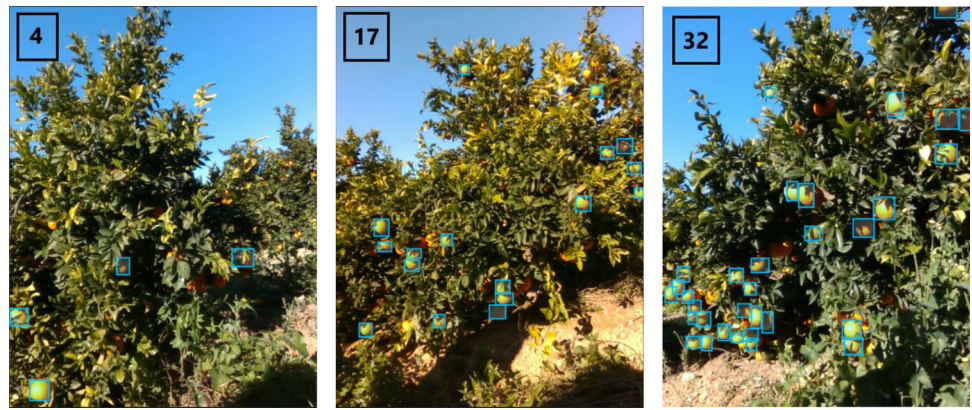


Fig. 5 Some examples of the noisy output of detection methods. The image on the left shows an example of YOLOv5 output; the one in the centre, an example of YOLOv7 output. The image on the right illustrates common defects

This method leverages the very high overlap among neighbouring frames of a video segment centred on a target tree. Multiple frame-wise measurements $\hat{c}_{ij}$ (weak estimates) of the number of visible oranges on the target tree are expected to lead to a more accurate estimate $\hat{c}_i$ of the tree's visible load and to provide a useful characterisation of the model uncertainty. It is supported by a principle of machine learning: fusing weak pieces of evidence (estimates) can lead to a robust estimate. This fusion scheme should also provide robustness to inaccuracies in tree detection (across video segments) and to complexities common in farming scenarios such as high fruit densities, variable lighting conditions, and partial fruit overlaps.

In this work, m will be a regression DNN and S , the sample median in order to mitigate the effect of outliers (very deviated predictions). Outliers are mainly expected when estimating values in both tails of the annotation distribution (see Fig. 3). The low density in these regions could limit model generalisation.

3.4 Performance metrics

Model performance will be measured through accuracy and uncertainty metrics. Accuracy will be obtained in terms of the disagreement between estimates and expectations, and uncertainty, by measuring the spread of multiple estimates.

Accuracy will be calculated both at frame level and at sequence level. Let $y_i = \{c_{ij}\}$ be the series of annotations corresponding to a sequence/tree x_i , such that c_{ij} is the annotated count of the frame $f_{ij} \in x_i$. For the purposes of training and evaluating a DNN, both f_{ij} and x_i are labelled with c_i , calculated as the average of frame manual annotations c_{ij} (no outlier is expected):

$$c_i = \frac{1}{l} \sum_{j=0}^{l-1} c_{ij}, \quad (3)$$

That is, for learning purposes, frames will not be related to their individual annotations, but to an average annotation c_i shared by x_i all f_{ij} . This decision is expected to contribute

to model generalisation, in the same way that diverse samples share the same category in a classification task.

Accuracy is calculated in terms of both the median absolute error (MedAE) and the median absolute percentage error (MedAPE). At frame level, they are denoted as MedAE_f and as MedAPE_f , respectively, and formulated as

$$\text{MedAE}_f = \text{median}\{|c_i - \hat{c}_{ij}|\} \quad 0 \leq i < n, \quad 0 \leq j < l \quad (4)$$

$$\text{MedAPE}_f = \text{median}\left\{\left|\frac{c_i - \hat{c}_{ij}}{c_i}\right|\right\} \quad 0 \leq i < n, \quad 0 \leq j < l \quad (5)$$

with n being the total number of sequences/trees. At sequence/tree level, these two measures are denoted as MedAE_s and MedAPE_s and defined as:

$$\text{MedAE}_s = \text{median}\{|c_i - \hat{c}_i|\} \quad 0 \leq i < n \quad (6)$$

$$\text{MedAPE}_s = \text{median}\left\{\left|\frac{c_i - \hat{c}_i}{c_i}\right|\right\} \quad 0 \leq i < n \quad (7)$$

Uncertainty is computed at frame level for each tree in terms of the Median Absolute Deviation (MAD). MAD is defined as the median of the absolute deviations from the sample's median, and it is accepted as a robust measure of the sample variability. Given a set of estimates $\hat{y}_i = \{\hat{c}_{ij}\}$ from frames representing t_i , the estimate uncertainty MAD_i at tree i is computed as

$$\text{MAD}_i = \text{median}\{|\hat{c}_{ij} - \hat{c}_i|\} \quad 0 \leq j < l \quad (8)$$

A global MAD is also computed over all trees, considering the absolute deviation of every frame prediction $\hat{c}_{ij}$ from its tree prediction $\hat{c}_i$:

$$\text{MAD} = \text{median}\{|\hat{c}_{ij} - \hat{c}_i|\} \quad 0 \leq i < n, \quad 0 \leq j < l \quad (9)$$

3.5 Regression models

Two DNN models were adapted for regression: EfficientNetB0 [20] and feature pyramid network [19] (FPN) using the former model as the backbone to make results comparable. EfficientNetB0 is a convolutional neural network resulting from a scaling method that optimises depth, width, and resolution jointly until reaching a light and balanced architecture. FPN combines low-resolution maps with strong semantic information, with high-resolution maps with weak semantic information, on the so-called top-down pathway. It was originally designed for multi-scale object detection, although it is reshaped here to cope with regression.

Table 3 EfficientNetB0 adapted as FPN backbone. Summary of the network architecture for input images of size (480, 640)

Block	Operator	Output resolution	Lateral connection
stem	Conv3x3	(240, 320)	
1	MBCConv1, k3x3	(240, 320)	
2	MBCConv6, k3x3	(120, 160)	P2
3	MBCConv6, k5x5	(60, 80)	P3
4	MBCConv6, k3x3	(30, 40)	
5	MBCConv6, k5x5	(30, 40)	P4
6	MBCConv6, k5x5	(15, 20)	
7	MBCConv6, k3x3	(15, 20)	P5
FCN	GlobalAveragePooling + batch normalisation + dropout Dense + batch normalisation + dropout Output unit		

Table 3 summarises the EfficientNetB0 model and identifies the lateral connections used by FPN to inject spatial details into multi-scale semantic maps. When used alone, the convolutional base of EfficientNetB0 is connected to a small fully connected network (FCN) as described at the bottom of the table.

It is expected that combining FPN multi-scale maps will result in better opportunities to represent oranges of different sizes. Two types of feature combination schemes are proposed. The first one combines maps after linearising them. The second one combines maps after upscaling them to a common resolution; thus, it considers the spatial distribution of concepts. They will be denoted as FPN-linear and FPN-map, respectively. Figures 6 and 7 show both schemes.

4 Results

4.1 Experimental design

As introduced in Sec. 3.1, experiments involve four crop rows videos. Error estimation relied on a fourfold cross-validation, with the number of partitions (four) determined by the natural row distribution. This scheme guarantees a realistic scenario, since the particular acquisition conditions of each row (e.g. lighting conditions) remain isolated in the training or in the test subset, but not in both.

For tuning regression models, hyperparameters domains were defined. They involve architecture pieces such as dropout rates in $\{0.2, 0.3, 0.4, 0.5\}$ and dense layer sizes in $\{8, 16, 32, 64\}$, as well as learning factors such as the learning rate in $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$, the optimiser in $\{\text{Adam}, \text{Adam}+\text{CosineDecay}\}$, the output activation

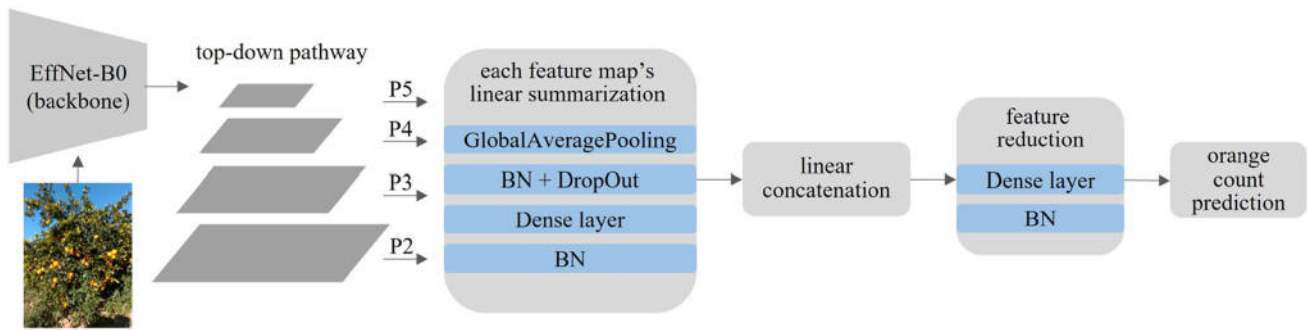


Fig. 6 FPN-linear: linearised maps are fused

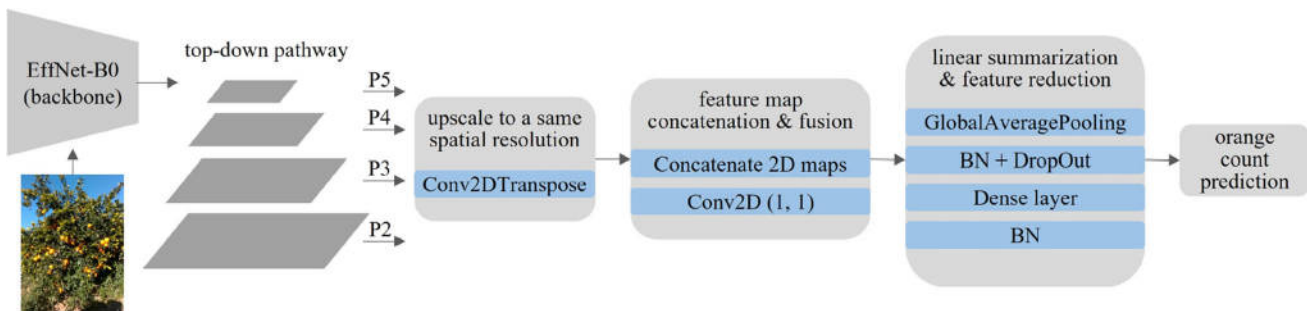


Fig. 7 FPN-map: upscaled 2D maps are fused

function in {linear, sigmoid}, and the batch size in {16, 24, 32}.

The training of both EfficientNetB0 and FPN includes the last convolutional block of the backbone (block 7 in Table 3). This allows the retraining of the most semantic features to make them more useful/specific for the regression task.

The number of experiments carried out was 960, each consisting of the training and evaluation of a regression model. This figure comes from combining 3 models, 4 sequence sizes, 4 data folds, and 20 random hyperparameter combinations.

Experiments were organised into two studies. The first one focuses on the analysis of multi-frame count by regression using three DNN on sequences of different lengths. In the second study, correlation analyses between outputs of two well-known detection models, YOLOv5 and YOLOv7, and manual annotations were conducted. Assuming that the former are automatic useful approximations, the intention is to judge the validity of the latter (used in training regression models).

4.2 Study I: counting by regression

A first analysis focuses on regression methods taking into account the two types of errors: MedAE and MedAPE.

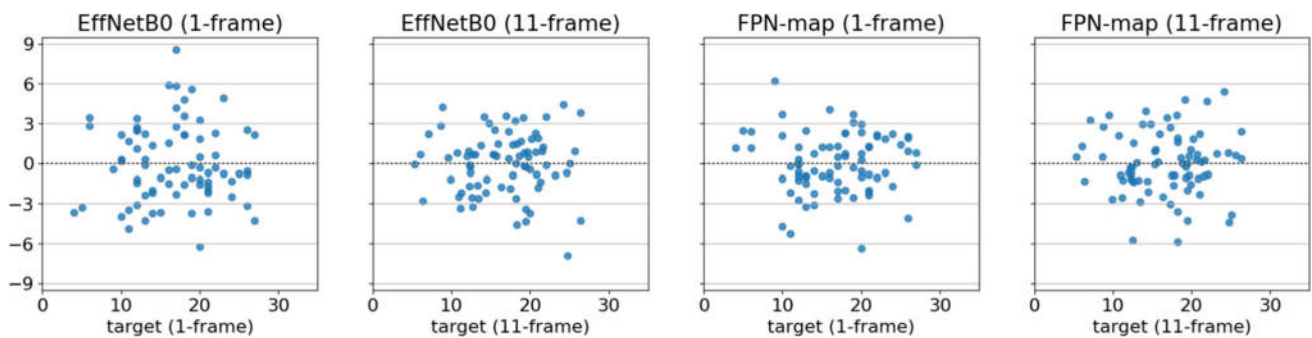
Two main issues are expected to be resolved: (1) Do sequences of frames provide value over individual frames? (2) Does the FPN model provide value over EfficientNetB0?

Table 4 summarises the error metrics of the regression models for different sequence lengths. Sequences of size 3, 7, and 11 frames include tree-level results, with the aim of visualising the effect of the sequence length on the estimation error. As can be seen, the median of the absolute error tends to decrease with the increase in the sequence length, and in the model complexity. In the cases of the EfficientNetB0 and FPN-map models, using 11-frame sequences reduces the MedAE (compared to using 1 frame) by 22% and 25%, respectively.

To gain a deeper knowledge about model behaviour, Fig. 8 shows the distributions of residuals (difference between target values and predictions) versus the target variable (orange count) for the EfficientNetB0 and FPN-map when processing 1-frame and 11-frame sequences (tree level). These cases summarise opposite scenarios. Residuals seem to follow almost random patterns, particularly in the range [10, 25], which concentrates higher densities. The greatest deviations occur at the extremes of the target variable domain, where the low density of samples could be affecting model generalisation. Outside that range, most FPN-map residuals are positive. That is, in regions with low data density, FPN-map underestimates the target value. Another observed trend is the reduction in

Table 4 Regression errors with respect to manual annotations for different sequence lengths (1, 3, 7, 11). MedAE is calculated at frame and sequence level: $\text{MedAE}_f/\text{MedAE}_s$. MedAPE is only computed at sequence level: MedAPE_s

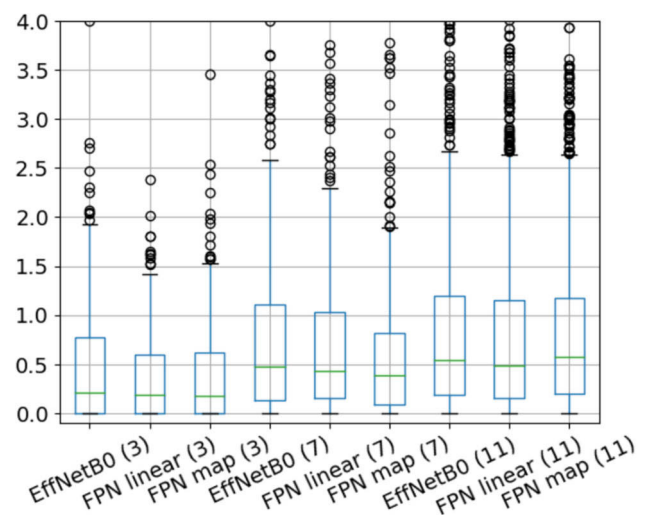
Model	1-frame		3-frame sequence		7-frame sequence		11-frame sequence	
	MedAE	MedAPE	MedAE	MedAPE	MedAE	MedAPE	MedAE	MedAPE
EffNetB0	3.20	0.186	2.70/2.71	0.169	2.61/2.66	0.155	2.61/2.51	0.151
FPN-linear	2.79	0.147	2.54/2.42	0.149	2.61/2.58	0.152	2.57/2.42	0.163
FPN-map	3.10	0.217	2.45/2.39	0.147	2.52/2.52	0.137	2.38/2.31	0.151

**Fig. 8** Distribution of regression residuals

large deviations in sequence-based solutions with respect to those based on individual frames.

Uncertainties of individual estimates were calculated by the median absolute deviations (MAD), as formulated in Eq. 9. Table 5 comprises the MAD values by model and sequence size. As expected, the greater the number of measurements (sequence length), the greater the dispersion with respect to central values. When models are compared, except for the case with 11-frame sequences, FPN-based models show lower uncertainty than the simpler EfficientNetB0 model. Figure 9 depicts the uncertainty distributions for each combination of regression method and sequence length. The effect of FPN in reducing uncertainty with respect to EfficientNet is clearly observed for lengths 3 and 7.

Finally, in order to statistically establish the value of using multi-frame count estimation, all paired series of tree-level MedAE_s from 1-frame sequences and l -frame sequences (for $l \in \{3, 7, 11\}$) were compared by the Wilcoxon signed-rank test [47], a nonparametric statistical hypothesis test (it makes no assumption on data

**Fig. 9** Uncertainty distributions for each pair of method and sequence length

distribution). Table 6 summarises the resulting p -values. With the exception of FPN-linear, which performed similarly across all sequence sizes, count estimates from

Table 5 Uncertainties based on MAD calculation

Model	3-frame sequence MAD	7-frame sequence MAD	11-frame sequence MAD
EffNetB0	0.21	0.47	0.54
FPN-linear	0.19	0.43	0.48
FPN-map	0.17	0.39	0.57

Table 6 Results (p -values) of the Wilcoxon signed-rank tests when assessing statistical differences between paired series of MedAE_s from 1-frame sequences and l -frame sequences (for $l \in \{3, 7, 11\}$). Significant p -values are highlighted ($\alpha = 0.05$)

Model	1-fr. versus 3-fr. sequence p value	1-fr. versus 7-fr. sequence p value	1-fr. versus 11-fr. sequence p value
EffNetB0	0.014	0.023	0.005
FPN-linear	0.975	0.951	0.635
FPN-map	0.012	0.023	0.019

multiple frames were significantly more accurate than those based on a single frame (considering a significance level $\alpha = 0.05$). Test implementations provided by the Python library SciPy were used.

4.3 Study II: validating manual annotations

As introduced in Sec. 3.2.1, correlations between post-processed output of detection methods and manual annotations were calculated.

Table 7 shows Pearson's and Spearman's correlation coefficients. While the former measures linear dependence, the latter measures monotonic dependence (linear or not). Very high positive correlations were observed between the YOLOv5 and YOLOv7 outputs and manual annotations, which provides evidence in favour of the usefulness of the latter. These correlations grew along with sequence length, confirming the value of creating annotations based on multiple frames (see Eq. 3). They were similar to those found between the outputs of YOLOv5 and YOLOv7. All p -values were virtually zero.

5 Discussion

The problem of counting by regression the number of visible oranges in trees from crop row videos has been studied. This is a real-world complex scenario affected by a number of conditions that make data annotation, model learning, and model evaluation challenging. As has been proved in previous studies [48], under heterogeneous

conditions in terms of fruit density, fruit colour, fruit size, and light exposure, performances of machine learning solutions drop significantly.

In this study, acquisition conditions of each crop row video were unique, since they were captured at different times and at various locations in the field. A second factor was the wide range of orange tones observed, going from very soft and bright colours close to white, in the presence of high solar radiation, to very dark colours in regions with a lot of shadows. Third, overlaps and shadows produced by orange clusters or by branches and leaves were also very frequent. In addition, data partitions respected the natural distribution of trees in rows, so different acquisition conditions were not shared between training and testing phases. This reinforced realistic conditions and allowed unbiased measurements of model generalisation. This data use pattern is recommended for future research. All of the above contributed to creating an inherently complex regression task.

Data annotation (assigning each frame the number of visible oranges) became a first challenge. How much visual evidence should be considered to count an orange? How much human experience is involved in identifying an orange? How much visual evidence is necessary for the regression model to learn an effective mapping? After evaluating several alternatives, a conservative manual labelling strategy was chosen aimed at minimising subjective decisions, although perhaps to the detriment of the number of false negatives. The result was successfully validated by calculating correlations with the outputs of pretrained detection methods (YOLOv5 and YOLOv7), which were considered rough (noisy) approximations of the

Table 7 Correlations between the output of detection methods and manual annotations. Very high positive coefficients with p -values virtually zero. Correlation increases with the sequence length. Results suggest the validity of manual annotations

Model	1-frame		3-frame sequence		7-frame sequence		11-frame sequence	
	Pearson	Spearman	Pearson	Spearman	Pearson	Spearman	Pearson	Spearman
YOLOv5-Annot	0.82	0.81	0.88	0.86	0.90	0.89	0.91	0.90
YOLOv7-Annot	0.81	0.80	0.85	0.84	0.87	0.87	0.87	0.88
YOLOv5-YOLOv7	0.87	0.88	0.90	0.89	0.91	0.90	0.91	0.91

number of visible oranges. Future efforts could explore hybrid strategies, in which the outputs of pretrained detection methods can be manually corrected. However, as discussed above, the number of visible fruits on a tree is not the ultimate goal, but only a means (a sample of the unknown fruit load) to achieve an end (estimation of actual production).

Regression solutions for fruit counting in more controlled and simpler tasks have shown error rates around 9% [38]. Therefore, the MedAPE of 13.7% shown by FPN-map for 7-frame sequences can be considered a promising rate in a complex scenario.

Results were obtained from a specific acquisition configuration, including the frame rate (30 FPS), the FOV (about 2 ms wide by 2.5 ms high), and the farm vehicle speed (3–5 km/h). At this speed, the row segment occupied by each tree would have been described by at least 30 frames. Considering that tree-centric 7-frame sequences led to the best results, the proposed solution should be robust to increases in vehicle speed and decreases in frame rate, and even to different degrees of video subsampling. Regarding the FOV, considering that the estimated number of oranges is expected to be used as a sample of the unknown population of oranges in the tree, a significant reduction in the FOV should not have a significant impact on the ability of the estimates to characterise the relative yield distribution across the crop field, as long as the observable area in the image includes a significant portion of the tree canopy. The relationships between acquisition parameters and the performance of machine vision-based methods offer future research opportunities.

To date, estimating production in orange orchards still relies mainly on human observers by simple inspection of the visible fruits. They empirically relate observed distributions of visible fruits with physical magnitudes of product biomass (e.g. kg.) based on their experience over many years. Given the manual and subjective nature of this practice, it is considered inaccurate, time-consuming, tedious, non-scalable, and observer dependent. The ultimate goal of this research is to create an automatic and systematic observer which, based on objective measurements of the distribution of visible oranges along the crop rows, makes a robust and accurate estimate of the production of an orange orchard. This learning objective requires yield annotations from past harvests not yet available.

Meanwhile, the robust estimation of the number of visible fruits per tree could have a straightforward application. Assuming that the number of visible fruits on a tree is positively and significantly correlated with the total, unknown fruit load for most trees, it is already possible to estimate the relative frequency distribution of fruit load across the orchard. This information can support the

comparison of production between different crop areas (spatial distribution) or different seasons (temporal distribution), as well as the optimisation of resource assignment in farm activities.

6 Conclusions

This work explores the opportunities of counting the number of visible oranges on trees by direct regression, from crop row videos captured in real scenarios. To the best of our knowledge, no previous work has dealt with a similar use case.

Three regression models were compared (EfficientNetB0 and two others based on Feature Pyramid Networks) when computing count estimates from individual frames and from tree-centred sequences of frames. The feasibility of counting by regression the number of oranges from tree images in the wild was demonstrated. In particular, the benefit of frame sequences in calculating more accurate estimates was statistically established, as well as the advantages of combining FPN multi-scale maps. The lowest regression errors were obtained from sequences of 7 and 11 frames, achieving reductions of 25% of the error calculated from individual frames.

A new annotated database was also created. It is composed of crop row videos together with annotations (number of visible oranges) of a set of relevant tree-related frames. It is now available to the scientific community.

Future extensions of these ideas could consider new annotation strategies, including more practical (although difficult) ways of measuring production such as the total weight in kg of fruits on the tree. Solutions that process entire video sequences, rather than combining frame-level decisions, could also be evaluated.

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Data availability The Ambimetrics Orange Counting Dataset introduced in the Dataset subsection and the data annotations described in

the Data annotation subsection have been created as part of this research. All experiments performed and results obtained in this work are supported by these data. The dataset is publicly available and can be downloaded as a ZIP file for free from the website <https://smart-cai.ambimetrics.com>. Data annotations are provided in both spreadsheet (Microsoft excel) and CSV formats. Ambimetrics S.L. has copyright on all collected data.

Declarations

Conflict of interest The authors declare that they have no conflict of interest that could affect the independence of the research reported in this paper.

Ethical approval Not applicable.

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